

The first-passage time density for the diffusion model with variable drift and variable starting point

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Abstract

The Ratcliff diffusion model is fitted with across-trial variability in drift rate, starting point, and non-decision time. The drift integral of the first-passage time density has been available in closed form; the starting-point integral has been evaluated numerically in every implementation, and Tuerlinckx (2004) concluded that no closed form exists. We show that this holds only for the large-time series representation. In the small-time representation the starting-point integral is a Gaussian moment, and we give the density with normally distributed drift and uniformly distributed starting point in closed form, together with a large-time counterpart in terms of the Faddeeva function. All partial derivatives are closed form, and the seven-parameter likelihood needs a single one-dimensional quadrature. The result is verified symbolically, to 48 digits, against three reference implementations and process simulation, and is machine-checked in Lean 4.

Keywords: Diffusion model, Response time modeling, First-passage time, Starting point variability

1. Background

The diffusion model for response times (Ratcliff, 1978) describes a two-choice decision as a Wiener process with drift v and unit diffusion coefficient¹ evolving between two absorbing barriers at 0 and $a > 0$ from a starting point aw , $0 < w < 1$. The first-passage times at the two barriers are the predicted response times for the two alternatives. To account for the fast and slow errors observed in most response time experiments, the model is applied with across-trial variability in three of its quantities (Ratcliff, 1978; Ratcliff and Rouder, 1998; Ratcliff and Tuerlinckx, 2002): the drift rate is normally distributed, $v \sim N(v, \eta^2)$; the starting point is uniformly distributed, $w \sim U(w_1, w_2)$, with Ratcliff's $s_z = a(w_2 - w_1)$; and the non-decision time is uniformly distributed over a window of width s_t . The predicted density of a response time is then a triple integral of the constant-parameter first-passage time density over these three distributions.

Of the three integrals, only the drift integral has been available in closed form. The first-passage time density with normally distributed drift is given by Horrocks and Thompson (2004) and, for the lower barrier, reads (Blurton et al., 2017, Eq. 1)

$$g(t | v, \eta^2, a, w) = \frac{1}{\sqrt{2\pi}^3 S} \exp\left[\frac{-v^2 t - 2vaw + \eta^2(aw)^2}{2S}\right] \times \sum_{j=0}^{\infty} (-1)^j r_j \exp\left(-\frac{r_j^2}{2t}\right), \quad (1)$$

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¹For diffusion coefficient $\sigma^2 \neq 1$ all results below hold after the substitution $(v, \eta^2, a) \rightarrow (v/\sigma, \eta^2/\sigma^2, a/\sigma)$ (Blurton et al., 2017).

where $S := 1 + \eta^2 t$, $r_j = ja + aw$ for even j and $r_j = ja + a(1 - w)$ for odd j . Blurton et al. (2017) derived the corresponding cumulative distribution function. Both results are stated in the small-time representation of the first-passage time series (Navarro and Fuss, 2009; Gondan et al., 2014), that is, as a sum over image charges r_j with Gaussian factors $\exp(-r_j^2/2t)$.

The starting-point and non-decision-time integrals have remained numerical. Tuerlinckx (2004, pp. 706–708) worked in the large-time representation of Ratcliff (1978), in which the starting point enters through $\sin(k\pi w)$, integrated the starting point analytically at fixed drift, and concluded that the drift integral then has to be approximated by quadrature. Every current implementation of the seven-parameter model integrates the starting point numerically: DMAT (Vandekerckhove and Tuerlinckx, 2008), fast-dm and rtdists (Voss and Voss, 2007; Singmann et al., 2022), HDDM (Wiecki et al., 2013), WienR (Hartmann and Klauer, 2021), and the Stan implementation of Henrich et al. (2024). Blurton et al. (2017) noted that adding starting-point and non-decision-time variability to their result “requires numerical evaluation of two integrals” (p. 11).

In this note we show that the starting-point integral of the density (1) has a closed form. The reason it was missed is representational. In the large-time series, after the drift has been integrated, the starting point appears in a Gaussian factor with a *positive* coefficient of w^2 and the integral against $\sin(k\pi w)$ lands on the imaginary error function; this is the obstruction Tuerlinckx encountered. In the small-time series the factor $\exp(-r_j^2/2t)$ contributes a negative coefficient of w^2 that always dominates the positive one produced by the drift integral, so the integrand is a linear function times a decaying Gaussian, and the integral is a Gaussian moment with a closed form. Section 2 gives the result and its derivation, Section 3 the large-time

counterpart that is needed for numerical stability at large t , Section 4 the truncation rule, Section 5 the partial derivatives, Section 6 the resulting seven-parameter likelihood, and Section 7 the numerical verification and a comparison of computational cost with existing software.

2. The density with variable drift and variable starting point

Let the starting point be uniformly distributed on $[w_1, w_2]$, $0 \leq w_1 < w_2 \leq 1$. The lower-barrier first-passage time density of the resulting process is

$$g(t | \nu, \eta^2, a, w_1, w_2) := \frac{1}{w_2 - w_1} \int_{w_1}^{w_2} g(t | \nu, \eta^2, a, w) dw. \quad (2)$$

Because $r_j \geq ja$ for $w \in [0, 1]$, the j -th term of (1) is bounded in absolute value by $(j+1)a \exp(-j^2 a^2/2t)$ times a constant independent of w and j (Appendix B), so the series converges absolutely and uniformly on $[w_1, w_2]$ and summation and integration may be exchanged. It therefore suffices to integrate one term.

2.1. The j -th term is a Gaussian moment

Write $r_j = p_j + q_j w$ with $p_j = ja$, $q_j = a$ for even j , and $p_j = (j+1)a$, $q_j = -a$ for odd j . Collecting the w -dependence of the j -th term of (1), that is, of $\exp[(-2\nu aw + \eta^2 a^2 w^2)/2S] r_j \exp(-r_j^2/2t)$, gives a quadratic exponent in w ,

$$(p_j + q_j w) \exp\left(-\frac{1}{2}Aw^2 + B_j w + C_j\right), \quad (3)$$

with

$$A = \frac{a^2}{tS}, \quad B_j = -\frac{\nu a}{S} \mp \frac{k_j a^2}{t}, \quad C_j = -\frac{\nu^2 t}{2S} - \frac{k_j^2 a^2}{2t}, \quad (4)$$

where $k_j = j$ and the upper sign apply for even j , and $k_j = j+1$ and the lower sign apply for odd j . The coefficient of w^2 deserves attention. The drift integral contributes $+\eta^2 a^2/2S$, a growing Gaussian in w , whereas the image-charge factor contributes $-a^2/2t$, and

$$\frac{a^2}{2t} - \frac{\eta^2 a^2}{2S} = \frac{a^2(S - \eta^2 t)}{2tS} = \frac{a^2}{2tS} > 0 \quad (5)$$

for every $t > 0$, every η , and every j . The combined exponent is therefore a decaying Gaussian with $A > 0$. This is the whole content of the result: in the small-time representation the starting-point integrand is a linear function times a proper Gaussian, and such integrals have closed forms in terms of $\exp$ and Φ .

2.2. The result

For $A > 0$ and $m := B/A$,

$$\begin{aligned} & \int_{w_1}^{w_2} (p + qw) \exp\left(-\frac{1}{2}Aw^2 + Bw + C\right) dw \\ &= (p + qm) \sqrt{\frac{2\pi}{A}} \exp\left(C + B^2/2A\right) \\ & \quad \times \left[\Phi\left(\sqrt{A}(w_2 - m)\right) - \Phi\left(\sqrt{A}(w_1 - m)\right)\right] \\ & \quad - \frac{q}{A} \left[\exp\left(C + Bw_2 - \frac{1}{2}Aw_2^2\right) \right. \\ & \quad \left. - \exp\left(C + Bw_1 - \frac{1}{2}Aw_1^2\right)\right], \end{aligned} \quad (6)$$

where Φ denotes the standard normal distribution function (Appendix A). Applying (6) term by term with the coefficients (4) gives the lower-barrier first-passage time density with normally distributed drift and uniformly distributed starting point,

$$\begin{aligned} g(t | \nu, \eta^2, a, w_1, w_2) &= \frac{1}{(w_2 - w_1) \sqrt{2\pi t^3 S}} \sum_{j=0}^{\infty} (-1)^j I_j, \quad (7) \\ I_j &= \int_{w_1}^{w_2} (p_j + q_j w) \\ & \quad \times \exp\left(-\frac{1}{2}Aw^2 + B_j w + C_j\right) dw, \end{aligned}$$

with I_j given in closed form by (6). The density at the upper barrier is $g(t | -\nu, \eta^2, a, 1 - w_2, 1 - w_1)$. For $w_2 - w_1 \rightarrow 0$ the result reduces to (1), and for $\eta = 0$ it reduces to the starting-point integral of the constant-drift density, so it can be used in a fitting routine regardless of whether either form of variability is present in the data.

Two remarks on evaluation. First, for large j the center $m = B_j/A$ of the Gaussian in (6) lies far outside $[w_1, w_2]$, so that $\exp(C + B^2/2A)$ is very large and the bracketed difference of Φ values is very small. The product must be formed in logarithmic space, using $\log \Phi$ and a tail-aware difference; with that precaution the expression is stable for all parameter values we have tested. Second, the identity $\sqrt{2\pi} \exp(C + B^2/2A) \phi(\sqrt{A}(w_i - m)) = \exp(C + Bw_i - \frac{1}{2}Aw_i^2)$, where ϕ is the standard normal density, shows that the two brackets in (6) share their exponential factors; this is used in the derivatives of Section 5.

3. Large-time representation

Like every small-time series, (7) loses precision through cancellation when t/a^2 is large; in double precision the loss becomes visible for $t/a^2 \gtrsim 2$. A large-time counterpart is obtained from the large-time series of Ratcliff (1978). Integrating $\exp(-\nu aw - \nu^2 t/2)$ against the $N(\nu, \eta^2)$ drift density produces exactly the prefactor of (1), so the large-time density with normally distributed drift is

$$\begin{aligned} g(t | \nu, \eta^2, a, w) &= \frac{\pi}{a^2 \sqrt{S}} \exp\left[\frac{-\nu^2 t - 2\nu aw + \eta^2 (aw)^2}{2S}\right] \\ & \quad \times \sum_{k=1}^{\infty} k \exp(-k^2 \pi^2 t/2a^2) \sin(k\pi w). \end{aligned} \quad (8)$$

Here there is no image-charge factor to compensate the growing Gaussian, and the starting-point integral of the k -th term is $\int_{w_1}^{w_2} \exp(\kappa w^2 + \lambda w) \sin(k\pi w) dw$ with $\kappa = \eta^2 a^2 / 2S > 0$ and $\lambda = -va/S$. Written as the imaginary part of $\int \exp(\kappa w^2 + \mu_k w) dw$ with $\mu_k = \lambda + ik\pi$, the antiderivative is an imaginary error function of complex argument, which overflows. It is, however, bounded when expressed through the Faddeeva function $w(z) = \exp(-z^2) \operatorname{erfc}(-iz)$ (Abramowitz and Stegun, 1964, 7.1.3):

$$\int_{w_1}^{w_2} \exp(\kappa w^2 + \mu_k w) dw = \frac{-i\sqrt{\pi}}{2\sqrt{\kappa}} \left[\exp(\kappa w^2 + \mu_k w) w(z_k(w)) \right]_{w_1}^{w_2}, \quad (9)$$

with $z_k(w) = \sqrt{\kappa} w + \mu_k / 2\sqrt{\kappa}$. Since $\operatorname{Im} z_k = k\pi / 2\sqrt{\kappa} > 0$, $w(z_k)$ lies in the upper half plane where it is bounded by one, and the product with $\exp(\kappa w^2 + \mu_k w)$ is of the size of the integrand. For $\eta = 0$ the bracket reduces to $(\exp(\mu_k w_2) - \exp(\mu_k w_1)) / \mu_k$. The large-time density with uniformly distributed starting point is then

$$g(t | v, \eta^2, a, w_1, w_2) = \frac{\pi \exp(-v^2 t / 2S)}{a^2 \sqrt{S} (w_2 - w_1)} \sum_{k=1}^{\infty} k \exp(-k^2 \pi^2 t / 2a^2) \times \operatorname{Im} \int_{w_1}^{w_2} \exp(\kappa w^2 + \mu_k w) dw. \quad (10)$$

Fast and accurate algorithms for $w(z)$ are standard (Zaghloul and Ali, 2011) and available in scientific libraries. In our implementation the two representations are switched at $t/a^2 = 1$; they agree to 1.5×10^{-11} relative across the switch. We stress that (9) must be used in the Faddeeva form: written with the imaginary error function it loses of the order of $\pi^2 k^2 / 4\kappa$ nats to cancellation and returns meaningless values when the series is truncated at a fixed large K .

4. Convergence

The j -th term of (7) is bounded in absolute value by $c(j+1)a \exp(-j^2 a^2 / 2t)$, and the k -th term of the large-time series (10) by $c' k \exp(-k^2 \pi^2 t / 2a^2)$, where the constants c and c' do not depend on the index (Appendix B). Both bounds decrease super-exponentially, so the truncation error is of the order of the first neglected term, and for a tolerance ε the series may be cut at

$$J = \left\lceil \frac{\sqrt{2t \log(1/\varepsilon)}}{a} \right\rceil + 1, \quad K = \left\lceil a \sqrt{\frac{2 \log(1/\varepsilon)}{\pi^2 t}} \right\rceil + 1. \quad (11)$$

With $\varepsilon = 10^{-12}$ the small-time series needs $J \leq 9$ terms for $t/a^2 \leq 1$, and the large-time series needs $K \leq 4$ terms for $t/a^2 \geq 1$. Evaluating (7) with $\varepsilon = 10^{-12}$ and with $\varepsilon = 10^{-40}$ on the 400 parameter sets of Section 7 gives results that differ by at most 2.2×10^{-16} .

5. Partial derivatives

Gradient information is required by Hamiltonian Monte Carlo and by gradient-based optimizers, and Hartmann and

Klauer (2021) derived it for the constant-parameter model. Every term of (7) and (10) is a composition of $\exp$, Φ , and w , so the partial derivatives with respect to v , η , a , w_1 , and w_2 are again closed form. For the small-time representation it is convenient to differentiate I_j with respect to its seven coefficients and to apply the chain rule. Writing $G := \sqrt{2\pi/A} \exp(C + B^2/2A)[\Phi(\sqrt{A}(w_2 - m)) - \Phi(\sqrt{A}(w_1 - m))]$ and $E_i := \exp(C + Bw_i - \frac{1}{2}Aw_i^2)$, so that $I = (p + qm)G - (q/A)(E_2 - E_1)$, the total differential of I under $(dA, dB, dC, dp, dq, dw_1, dw_2)$ is

$$\begin{aligned} dI &= (dp + m dq + q dm)G + (p + qm) dG \\ &\quad - \left(\frac{dq}{A} - \frac{q dA}{A^2} \right) (E_2 - E_1) - \frac{q}{A} (dE_2 - dE_1), \\ dm &= \frac{dB - m dA}{A}, \quad dK = dC + m dB - \frac{1}{2}m^2 dA, \\ dG &= G \left(dK - \frac{dA}{2A} \right) + E_2 \left[(dw_2 - dm) + \frac{(w_2 - m) dA}{2A} \right] \\ &\quad - E_1 \left[(dw_1 - dm) + \frac{(w_1 - m) dA}{2A} \right], \\ dE_i &= E_i \left[dC + w_i dB - \frac{1}{2}w_i^2 dA + (B - Aw_i) dw_i \right], \end{aligned} \quad (12)$$

in which the identity noted at the end of Section 2 has been used to express dG through E_1 and E_2 rather than through ϕ . The coefficient differentials follow from (4); for instance $\partial B_j / \partial v = -a/S$ and $\partial C_j / \partial v = -vt/S$, and A , p_j , q_j do not depend on v . For the large-time representation only two identities are needed, both immediate from $\frac{d}{dw} \exp(\kappa w^2 + \mu w) = (2\kappa w + \mu) \exp(\kappa w^2 + \mu w)$: with $I := \int_{w_1}^{w_2} \exp(\kappa w^2 + \mu w) dw$ and $\mathcal{E}_i := \exp(\kappa w_i^2 + \mu w_i)$,

$$\begin{aligned} \frac{\partial I}{\partial \mu} &= \frac{(\mathcal{E}_2 - \mathcal{E}_1) - \mu I}{2\kappa}, \\ \frac{\partial I}{\partial \kappa} &= \frac{(w_2 \mathcal{E}_2 - w_1 \mathcal{E}_1) - I - \mu \partial I / \partial \mu}{2\kappa}. \end{aligned} \quad (13)$$

The derivatives with respect to w_1 and w_2 are, by the fundamental theorem of calculus, the integrand evaluated at the endpoints, corrected for the factor $1/(w_2 - w_1)$. All of (12) and (13), and every chain-rule coefficient, are verified as exact symbolic identities in the supplementary code; the resulting gradients agree with those of Hartmann and Klauer (2021) as implemented in WienR to 3×10^{-13} absolute on the parameter grid of Section 7.

6. The seven-parameter likelihood

With the non-decision time uniformly distributed on $[t_0, t_0 + s_t]$, the density of an observed response time is

$$f(t) = \frac{1}{s_t} \int_{t-t_0-s_t}^{t-t_0} g(u | v, \eta^2, a, w_1, w_2) du, \quad (14)$$

where the lower limit is replaced by 0 if negative. This integral does not have a closed form: the non-decision time shifts the time argument, which enters every term of (7) through $\sqrt{t}$ and $1/t$, and we have not found a representation in which the shift integrates. What (7) achieves is to reduce the seven-parameter likelihood from a two-dimensional cubature over (w, u) to a

Table 1: Maximum relative error of Gauss–Legendre quadrature for (14) against adaptive quadrature, 200 random parameter sets, densities above 10^{-3} .

Nodes	Plain	With cubic map
8	4.1×10^{-3}	4.3×10^{-3}
16	7.3×10^{-5}	3.5×10^{-6}
24	4.1×10^{-6}	2.7×10^{-8}
32	1.1×10^{-8}	1.5×10^{-9}

one-dimensional quadrature over u of a smooth closed-form integrand on a short interval. Because $g(u)$ rises steeply from zero at $u = 0$ and the window may include that leading edge, a plain Gauss–Legendre rule converges slowly; the substitution $u = u_{lo} + (u_{hi} - u_{lo}) s^3$, which clusters nodes at the lower limit, fixes this. Table 1 gives the maximum relative error of the rule against adaptive quadrature over the 200 seven-parameter sets described below. A 32-node rule with the cubic map reaches 1.5×10^{-9} at a fixed cost of 32 evaluations of (7). The gradient of (14) with respect to the five parameters of g is the same rule applied to the gradient of g , and $\partial f / \partial t_0 = [g(u_{lo}) - g(u_{hi})] / s_t$ and $\partial f / \partial s_t = [g(u_{lo}) - f] / s_t$ follow from the endpoints.

7. Verification and efficiency

7.1. Verification

We checked the result in five independent ways. (i) Every algebraic step, in particular the antiderivative (6), the inequality (5), the coefficient table (4), the drift integral leading to (1), and all derivative formulae of Section 5, is verified as an exact symbolic identity in a computer algebra system, that is, as simplify($dF/dx - f$) = 0 rather than by numerical agreement. (ii) In 40- to 60-digit arithmetic, (7) agrees with a brute-force double integral of the constant-parameter density over (v, w) to 8.7×10^{-32} , and (10) agrees with quadrature of (1) to between 10^{-35} and 10^{-48} at $t/a^2 = 0.9, 1.5$, and 4.7 . (iii) On 400 random parameter sets with $a \in \{0.6, 1, 1.5, 2.5\}$, $v \in [-3, 4]$, $\eta \in \{0.5, 1, 2\}$, $w_2 - w_1 \in \{0.05, 0.2, 0.4\}$, and $t \in [0.05, 3]$, spanning densities from 5×10^{-20} to 1.6 , the switched representation agrees with WienR (Hartmann and Klauer, 2021) called at its tightest tolerance (10^{-12}) to a maximum absolute difference of 1.2×10^{-14} ; rtdists (Singmann et al., 2022), whose diffusion code is that of fast-dm, converges onto the same values as its precision setting is raised (7.6×10^{-4} , 1.1×10^{-5} , 2.5×10^{-8} at precision 3, 5, 8), as does the HDDM likelihood (Wiecki et al., 2013) as its integration tolerance is tightened. On 200 seven-parameter sets, (14) agrees with WienR to 2.5×10^{-9} , which is WienR’s own reported cubature error. (iv) An Euler–Maruyama simulation of the process with 400 000 trials, step 5×10^{-5} , and the Broadie–Glasserman–Kou barrier correction gives a lower-barrier response time histogram in 30 bins of 100 ms whose χ^2 against the binned density (7) is 21.6, with all standardized residuals below 1.9. (v) The derivation of (7), including the antiderivative, the coefficient table, the exchange of summation and integration, and the Faddeeva integral (9), is formalized and machine-checked in Lean 4 with Mathlib (de Moura and Ullrich, 2021; The mathlib Community, 2020); the development

contains no unproved statement and relies on no axioms beyond Lean’s standard ones. The starting representations (1) and (8) are taken as definitions there, as they are taken from the literature here.

7.2. Efficiency

Table 2 compares the cost of (7) and (14), compiled from a direct transcription of the formulae, with the three packages in common use: WienR, which evaluates the drift-integrated series by adaptive cubature over w ; rtdists, whose diffusion code is that of fast-dm and integrates over w and the non-decision window by adaptive Simpson’s rule; and the HDDM likelihood, which does the same with the Navarro–Fuss series. All timings are single-threaded on one machine,² for 10 000 response times at one parameter set, which is the situation in a likelihood evaluation.

The error columns need a reference scale. Blurton et al. (2017) recommend a truncation tolerance of 1.5×10^{-8} , the square root of single-precision machine epsilon, and Gondan et al. (2014) illustrate their term counts at 10^{-3} . Tuerlinckx (2004, Table 1) reports differences of 2×10^{-7} to 3×10^{-4} between his quadrature-based algorithm and the classical one. Among current packages, rtdists (fast-dm) defaults to a precision of 3, documented as roughly the number of accurate decimals; the HDDM likelihood truncates its series at 10^{-4} and integrates over the starting point and the non-decision window by adaptive Simpson’s rule with tolerance 10^{-8} and recursion depth 10; the Stan implementation of Henrich et al. (2024) fixes the density tolerance at 10^{-6} and defaults to 10^{-4} for the derivatives; WienR defaults to 10^{-12} . The closed form is exact to double precision, and its seven-parameter quadrature at 1.5×10^{-9} is below every one of these tolerances.

Two features of Table 2 are worth stating. First, the closed form is the cheapest of the four at every entry and the only one that is exact for six parameters. rtdists costs 1.3 times as much for six parameters and twice as much for seven, and deviates from the density by 0.5% and 0.9%. HDDM costs 1.8 and 5 times as much and agrees to 4×10^{-6} and 1.5×10^{-7} ; its series tolerance of 10^{-4} leaves a floor near 10^{-7} that tightening its integration settings does not remove. WienR reaches 10^{-12} at 6 and 110 times the cost. Because the cost of the closed form is fixed whereas that of the integrators is set by the accuracy asked of them, these ratios grow with the tolerance. Second, the gradient comes almost free: all five partial derivatives of (7) share the exponential factors of the density itself, and an uncompiled vectorized implementation returns the density together with all five derivatives in $1.7 \mu\text{s}$, whereas WienR requires one further cubature per derivative. For reference, WienR’s own closed form for drift variability alone (Blurton et al., 2017) evaluates in $0.26 \mu\text{s}$ on the same machine, so the closed form with starting-point variability costs 1.3 times the drift-only density.

²Apple M1 Pro; R 4.5.3 with WienR 0.3-15 and rtdists 0.11-6; Python 3.11 with NumPy 1.26.4, SciPy 1.13.1, Cython 3.1.4, and hddm-wfpt 0.1.4, the HDDM likelihood distributed as a standalone package.

Table 2: Cost per evaluation in microseconds, single thread, 10 000 response times at $a = 1.2$, $\nu = 1$, $w = 0.5$, $\eta = 1$, $w_2 - w_1 = 0.2$, $s_t = 0.2$. Each package at the settings its fitting routine uses: WienR at tolerance 10^{-12} , rtdists at precision 3, HDDM at series tolerance 10^{-4} with Simpson tolerance 10^{-8} and recursion depth 10. Error is the maximum relative deviation from (7) or (14) over the parameter sets of Section 7 (densities above 10^{-3}); for the closed form the seven-parameter entry is the quadrature error of (14) against adaptive integration.

	Six parameters		Seven parameters (+ s_t)	
	Error	μs	Error	μs
Closed form	–	0.33	1.5×10^{-9}	9.2
WienR	2.5×10^{-12}	2.1	1.5×10^{-9}	1040
rtdists	4.6×10^{-3}	0.43	9.2×10^{-3}	18
HDDM	3.9×10^{-6}	0.60	1.5×10^{-7}	46
Closed form + 5 gradients [†]	–	1.7		
WienR, 5 gradients	10^{-13}	15.7		

[†]Uncompiled vectorized implementation.

8. Discussion

We have shown that the first-passage time density of the diffusion model with normally distributed drift and uniformly distributed starting point has a closed form, and that the reason it was not found earlier is the choice of series representation: the small-time series carries a Gaussian factor in the starting point that the large-time series lacks, and that factor is exactly what gives the integral a closed form. The result completes the analytic treatment of the drift and starting-point integrals of the seven-parameter model and leaves only the non-decision-time integral, which is a one-dimensional quadrature of a smooth function on a short interval and for which we have given a fixed 32-node rule accurate to 10^{-9} .

For practice the consequences are threefold. The density is exact up to a truncation error that is controlled by (11); it is evaluated at a fixed cost that is below that of each existing package at the settings the package fits with; and its partial derivatives are closed form and nearly free, which is what hierarchical Bayesian and Hamiltonian Monte Carlo implementations (Wiecki et al., 2013; Henrich et al., 2024) need. The result is directly usable in any of the existing packages, which already contain the drift-integrated density (1) as a building block and need only replace the quadrature over w by (6).

The corresponding cumulative distribution function with both forms of variability does not follow by the same route. Starting from the drift-integrated distribution function of Blurton et al. (2017), half of the series terms have a coefficient of w^2 that is exactly zero after the drift integral and close through the integral $\int \exp(cw)\Phi(\alpha w + \beta)dw$, which has a closed form; the other half carry the coefficient $+2a^2\eta^2$, a growing Gaussian times Φ , and reduce to an integral of Owen- T type that has no closed form in terms of $\exp$, Φ , or Owen’s T . For maximum likelihood and Bayesian estimation the density suffices, and we leave the distribution function open.

CRedit authorship contribution statement

Kianté Fernandez: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no competing interests.

Data availability

All code, reference values, comparison scripts, and the Lean 4 formalization are available in the supplementary material and at [repository URL to be added on acceptance].

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Appendix A. The antiderivative

Let $A > 0$, $m = B/A$, and

$$F(w) := (p + qm) \sqrt{\frac{2\pi}{A}} \exp(C + B^2/2A) \Phi(\sqrt{A}(w - m)) - \frac{q}{A} \exp(C + Bw - \frac{1}{2}Aw^2).$$

Using $\Phi' = \phi$ and $-\frac{1}{2}A(w - m)^2 + C + B^2/2A = C + Bw - \frac{1}{2}Aw^2$,

$$\begin{aligned} F'(w) &= (p + qm) \sqrt{2\pi} \exp(C + B^2/2A) \phi(\sqrt{A}(w - m)) \\ &\quad - \frac{q}{A} (B - Aw) \exp(C + Bw - \frac{1}{2}Aw^2) \\ &= [(p + qm) - q(m - w)] \exp(C + Bw - \frac{1}{2}Aw^2) \\ &= (p + qw) \exp(C + Bw - \frac{1}{2}Aw^2), \end{aligned}$$

which is the integrand of (6). The definite integral (6) is $F(w_2) - F(w_1)$. The same computation is carried out symbolically in the supplementary code and formally in the Lean development.

Appendix B. Term bounds

For $w \in [0, 1]$, $ja \leq r_j \leq (j + 1)a$, and the prefactor of (1) is bounded by $\exp[(2|\nu|a + \eta^2 a^2)/2S]$. Hence the j -th term of the series in (1), and therefore $|I_j|/(w_2 - w_1)$ in (7), is at most $(j + 1)a \exp(-j^2 a^2/2t)$ times a constant independent of j and w .

Since $\sum_j (j+1) \exp(-j^2 c)$ converges for every $c > 0$, the series converges absolutely and uniformly in w , which justifies the exchange of summation and integration and gives the truncation rule for J in (11). For the large-time series (10), $|w(z)| \leq 1$ in the upper half plane and $|\exp(\kappa w^2 + \mu_k w)| \leq \exp(\kappa + |\lambda|)$ on $[0, 1]$, so the k -th term is at most a constant times $k \exp(-k^2 \pi^2 t / 2a^2)$, which gives the rule for K .

Appendix C. Supplementary material

The supplementary material contains (i) a vectorized implementation of (7), (10), (14), and all partial derivatives in Python, together with a compiled kernel used for Table 2; (ii) the symbolic verification script; (iii) the scripts and reference values for the comparisons with WienR, rtdists, and HDDM, and the simulation; and (iv) the Lean 4 formalization.

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